



Prakhar Chauhan

Roll No.: B23BB1032

B.Tech. (BSBE, Minor: AI)

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EDUCATION

Degree	Institute	CGPA/%	Year
B.Tech. (BSBE, Minor: AI)	IIT Jodhpur	7.52	2023–2027
Senior Secondary	RBSE Board	87.2%	2022
Secondary	RBSE Board	80.5%	2020

TECHNICAL SKILLS

- **Languages & Backend:** Python, C/C++, JavaScript, TypeScript, Node.js, Express.js, FastAPI, Django, GraphQL, RESTful APIs
- **Databases:** PostgreSQL, MongoDB, Redis — schema design, indexing, query optimization
- **System Design:** Event-driven architecture, Kafka, BullMQ, WebSockets, ABR streaming, microservices
- **DevOps & Tools:** Docker, Git, Linux (Arch), FFmpeg, MLflow
- **ML / Data:** PyTorch, scikit-learn, NumPy, Pandas, RAG pipelines, Google Gemini API

EXPERIENCE

RAG-Powered CSR Compliance Analysis Engine

Jan 2025 – Mar 2025

Backend Developer | Django, PostgreSQL, Google Gemini API, RAG, Docker

GitHub

- Designed a Django REST backend serving real-time audit data with optimized PostgreSQL schemas achieving sub-15 ms query latency on 10k+ document records.
- Integrated a Retrieval-Augmented Generation pipeline using Google Gemini for automated ESG compliance auditing and containerized deployment using Docker.

Clinical Decision Support System

Aug 2024 – Dec 2024

Software Developer | Python, Decision Trees, Boolean Rule Engines

IIT Jodhpur & AIIMS Jodhpur

- Converted NCCN oncology guidelines into a traversable decision-tree engine with clinicians at AIIMS Jodhpur.

PROJECTS

Spark — Video Streaming Platform

Systems Engineer | TypeScript, Node.js, FFmpeg, Kafka, Redis, WebAssembly, C++

GitHub

- Architected an event-driven backend using BullMQ + Kafka for scalable asynchronous video transcoding.
- Designed adaptive bitrate HLS streaming pipeline with FFmpeg and WebAssembly-based bandwidth prediction reducing rebuffering by 30%.
- Implemented Redis caching and real-time stream monitoring using WebSockets and Apache Flink.

YOLOv12-BiFPN: Architecture Optimization & Ablation Study

Computer Vision Researcher | PyTorch, YOLOv12, Python

- Re-engineered the YOLOv12 feature fusion neck using a custom BiFPN v2 with a high-resolution P2 layer for improved small object detection.
- Implemented dynamically learned weighted feature fusion with depthwise separable convolutions for computational efficiency.
- Achieved **41.14% mAP** on MS COCO 2017 with a **+2.0% improvement** over baseline while maintaining a compact 2.6M–3.0M parameter range.

PneumoOps — Thoracic Disease Detection

ML Engineer | PyTorch, ONNX Runtime, FastAPI, Docker

GitHub

- Built A/B testing inference system routing requests between PyTorch and ONNX Runtime serving backends for chest X-ray classification.
- Implemented KS-test based drift monitoring and deployed the application using Docker and Hugging Face Spaces.

OPEN SOURCE CONTRIBUTION

Ghostty — Terminal Emulator

2025

Contributor | Zig, Terminal Protocols

GitHub

- Implemented OSC 3008 escape sequence support in Ghostty and contributed reviewed production-level code.

COMPETITIVE PROGRAMMING

Codeforces: Max Rating **1609** | **LeetCode:** Max Rating **1735** (500+ problems, 80+ Hard)

POSITIONS OF RESPONSIBILITY

- Secretary, Bioscience & Bioengineering Department — IIT Jodhpur
- Associate, The Product Club — IIT Jodhpur
- Volunteer, Prometeo (Annual Techfest) — IIT Jodhpur

KEY COURSES TAKEN

Data Structures & Algorithms, Introduction to Machine Learning, Deep Learning, Introduction to Computer Science, ML-DL-Ops